

CYP1B1 analysis of unilateral primary newborn glaucoma in Saudi children.

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Nonsyndromic primary newborn glaucoma, the most severe form of primary congenital glaucoma, typically is bilateral and often the result of CYP1B1 mutations, particularly in certain consanguineous populations. Truly unilateral cases are uncommon and genetically not well studied. During a 9-year period, we tested 5 consecutive children with unilateral primary newborn glaucoma from Saudi Arabia, where CYP1B1 mutations are the cause for 91% of bilateral primary newborn glaucoma cases. None of these children with unilateral primary newborn glaucoma harbored CYP1B1 mutations, suggesting that in this population the pathogenesis of unilateral disease differs from that of bilateral disease.

An unusual presentation of glaucoma in a neonate with Rubinstein-Taybi syndrome.

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To report the occurrence of unilateral, neonatal-onset congenital glaucoma in a child with Rubinstein-Taybi Syndrome (RTS). A 15-day-old male with features of RTS was presented with an enlarged corneal diameter, corneal haze, and peripheral corneal vascularization of the left eye. Ultrasound biomicroscopy of his left eye revealed iris atrophy, iridocorneal adhesions, and iris adhesions to a partially absorbed cataractous lens. Genetic evaluation of the child and the parents revealed a novel de novo heterozygous pathogenic variant in exon 5 of the CREBBP gene (NM_004380.3:c.1390C>T). A diode laser cyclophotocoagulation was performed to control the IOP in the left eye. Unilateral neonatal-onset congenital glaucoma due to iridocorneal adhesions can be a rare presentation of Rubinstein-Taybi Syndrome.

Exome sequencing identifies novel genes underlying primary congenital glaucoma in the National Birth Defects Prevention Study.

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Primary congenital glaucoma (PCG) affects approximately 1 in 10,000 live born infants in the United States (U.S.). PCG has a autosomal recessive inheritance pattern, and variable expressivity and reduced penetrance have been reported. Likely causal variants in the most commonly mutated gene, CYP1B1, are less prevalent in the U.S., suggesting that alternative genes may contribute to the condition. This study utilized exome sequencing to investigate the genetic architecture of PCG in the U.S. and to identify novel genes and variants. We studied 37 family trios where infants had PCG and were part of the National Birth Defects Prevention Study (births 1997-2011), a U.S. multicenter study of birth defects. Samples underwent exome sequencing and sequence reads were aligned to the human reference sample (NCBI build 37/hg19). Variant filtration was conducted under de novo and Mendelian inheritance models using GEMINI. Among candidate variants, CYP1B1 was most represented (five trios, 13.5%). Twelve probands (32%) had potentially pathogenic variants in other genes not previously linked to PCG but important in eye development and/or to underlie Mendelian conditions with potential phenotypic overlap (e.g., CRYBB2, RXRA, GLI2). Variation in the genes identified in this population-based study may help to further explain the genetics of PCG.

Aicardi-Goutières Syndrome with Congenital Glaucoma Caused by Novel TREX1 Mutation.

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Aicardi-Goutières syndrome (AGS) is a rare genetic disorder characterized by microcephaly, white matter lesions, numerous intracranial calcifications, chilblain skin lesions and high levels of interferon- α (IFN- α) in the cerebrospinal fluid (CSF). However, ocular involvement is reported significantly less frequently. We present a case of a neonate with hypotrophy, microcephaly, frostbite-like skin lesions, thrombocytopenia, elevated liver enzymes and hepatosplenomegaly. Magnetic resonance imaging (MRI) of the brain showed multiple foci of calcification, white matter changes, cerebral atrophy, and atrophic dilatation of the ventricular system. The inflammatory parameters were not elevated, and the infectious etiology was excluded. Instead, elevated levels of IFN- α in the serum were detected. Based on the related clinical symptoms, imaging and test findings, the diagnosis of AGS was suspected. Genetic testing revealed two pathogenic mutations, c.490C>T and c.222del (novel mutation), in the three prime repair exonuclease 1 (TREX1) gene, confirming AGS type 1 (AGS1). An ophthalmologic examination of the child at 10 months of age revealed an impaired pupillary response to light, a corneal haze with Haab lines in the right eye (RE), pale optic nerve discs and neuropathy in both eyes (OU). The intraocular pressure (IOP) was 51 mmHg in the RE and 49 in the left eye (LE). The flash visual evoked potential (FVEP) showed prolonged P2 latencies of up to 125% in the LE and reduced amplitudes of up to approximately 10% OU. This girl was diagnosed with congenital glaucoma, and it was managed with a trabeculectomy with a basal iridectomy of OU, resulting in a reduction and stabilization in the IOP to 12 mmHg in the RE and 10 mmHg in the LE without any hypotensive eyedrops. We present the clinical characteristics, electrophysiological and imaging findings, as well as the genetic test results of a patient with AGS1. Our case contributes to the extended ophthalmic involvement of the pathogenic c.490C>T and c.222del mutations in TREX1.

Neurofibromatosis with congenital glaucoma and buphthalmos in a newborn.

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A newborn infant with unilateral eyelid swelling, proptosis, and glaucoma is described. The diagnosis of congenital neurofibromatosis was made on the basis of a positive family history and by inspection of the patient. It is suggested that this clinical problem is more common at birth than generally recognized. Clinicians should be aware that congenital glaucoma secondary to neurofibromatosis is a distinct entity that can manifest itself in the newborn period.

Congenital glaucoma detected by unilateral megalophthalmos on prenatal sonography.

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A 36-week male fetus was noted to have isolated right megalophthalmos on prenatal ultrasound and was found to have buphthalmos and congenital glaucoma at birth. Detection of congenital glaucoma at birth may be occasionally possible if abnormal orbit dimensions are noted on late prenatal sonographic

examination. Early neonatal intervention may improve the chances to retain vision. © 2016 Wiley Periodicals, Inc. J Clin Ultrasound 45:499-501, 2017.

[Aniridia, congenital glaucoma and white corneas in a newborn baby].

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We present the case of a newborn baby in whom we observed aniridia, congenital glaucoma and edematous corneas, without clearance of the cornea after glaucoma surgery, and in whom a keratoplasty was considered. The patient had no evidence of systemic diseases and no deletion of chromosome 11. It is important to perform a thorough ophthalmological and systemic exploration in newborn patients with aniridia due to other diseases that can be associated with it. We present the unusual case in which aniridia, congenital glaucoma and white corneas coexisted in a newborn baby. To our knowledge, such cases have only been reported twice before in the literature (Arch Soc Esp Oftalmol 2009; 84: 573-576).

A rare case of primary congenital glaucoma in combination with neurofibromatosis 1: a case report.

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Neurofibromatosis 1 (NF1) is a common disease that mainly affects the skin and peripheral nervous system, and is characterized by bony dysplasia. Primary congenital glaucoma (PCG) is a sight-threatening disease that can manifest as a prodrome of NF1, especially in newborn babies. We report a case of PCG with NF 1. A 1-month-old boy presented with an enlarged right eyeball. An increased IOP and typical glaucomatous optic neuropathy were found, on the initial physical examination, a clinical diagnosis of primary congenital glaucoma (PCG) was made and a trabeculectomy with mitomycin C (MMC) therapy was subsequently performed. Three year later, the boy again presented with an even larger right eye and a gradually expanding left one. In addition to typical glaucomatous optic neuropathy, the boy also had multiple café au lait patches all over his body, megacephaly (head circumference = 60 cm; body weight = 14 kg; height = 93 cm) and remarkable facial features included swollen, soft upper eyelids and a flat, broad nose sphenoid wing dysplasia, eyelid thickening, bony orbit enlargement were found. It is rare have both PCG and NF1, and PCG may be a prelude to NF1. Continuous follow-up should be advised and we should raise our awareness of the combined condition and to improve chances for an early diagnosis.

Newborn primary congenital glaucoma: histopathologic features of the anterior chamber filtration angle.

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We present histopathologic findings that have not been previously reported as associated with abnormalities of the anterior chamber angle underlying newborn primary congenital glaucoma as a distinct entity. The major histopathologic findings were partial absence and retrodisplacement of Schlemm's canal, hypoplasia of the trabecular meshwork, broad attachment of ciliary muscle to the meshwork, and anterior insertion of hypoplastic iris with the formation of a pseudomembrane. These

profound anatomic derangements explain why goniosurgery is frequently unsuccessful in cases of newborn glaucoma.

Corneal anomalies in newborn primary congenital glaucoma.

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This report describes the corneal pathology in an infant with newborn primary congenital glaucoma and discusses whether these findings could be due to a developmental anomaly. The corneal specimen of a 4-month-old infant with newborn primary congenital glaucoma and cloudy corneas who had undergone penetrating keratoplasty was evaluated by light and electron microscopy. Light microscopy showed a thinned epithelium, areas of thickened Bowman's layer (approximately 27 μm thick) interspersed with nuclei, and a thickened and disorganized stroma. Descemet's membrane was intact, and the endothelium was mildly attenuated. The corneal changes seen in this patient may be specific to primary congenital glaucoma and may contribute to the corneal clouding seen so frequently in these patients.
